In this equation, { *F*, *H*} is not a commutator; it is a Poisson bracket. But still, Eq. 4.20 is suspiciously similar to Eq. 4.19. In fact, there is a close parallel between commutators and Poisson brackets, and their algebraic properties are quite similar. For example, if *F* and *G* represent operators, both commutators and Poisson brackets change their sign when *F* and *G* are interchanged. Dirac discovered this, and realized that it represents an important structural connection between the mathematics of classical mechanics and that of quantum mechanics. The formal identification between commutators and Poisson brackets is

$$[\mathbf{F}, \mathbf{G}] \qquad \Longleftrightarrow \qquad i\hbar\{F, G\}. \qquad (4.21)$$

To facilitate comparison with Eq. 4.19, we can substitute the symbols **L** and **H** that we've been using in this section.

$$[\mathbf{L}, \mathbf{H}] \qquad \Longleftrightarrow \qquad i\hbar\{L, H\}. \qquad (4.22)$$

Let's try and make this identification as clear as possible. If we start with Eq. 4.19,

$$\frac{d\mathbf{L}}{dt} = -\frac{i}{\hbar}[\mathbf{L}, \mathbf{H}],$$

and then use the identification of Eq. 4.22 to write the classical analog, the result is

$$\frac{d\mathbf{L}}{dt} = -\frac{i}{\hbar}(i\hbar\{\mathbf{L}, \mathbf{H}\})$$

or

$$\frac{d\mathbf{L}}{dt} = \{\mathbf{L}, \mathbf{H}\},$$